

Lecture 1: Introduction to matrices

Matrices

A table of real numbers with n row and k columns $\Rightarrow B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1k} \\ b_{21} & b_{22} & \dots & b_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & \dots & \dots & b_{nk} \end{pmatrix} \in \mathbb{R}^{n \times k}$

Notation

$\uparrow$ $B[i,j] = b_{ij}$ element in i th row and j th column

Operations:

- Addition: Only defined for matrices of the same size.

(element-wise): $(A + B)[i,j] = A[i,j] + B[i,j]$.

- Multiplication by scalar: Defined for any $\lambda \in \mathbb{R}$ and $A \in \mathbb{R}^{n \times k}$:

(element-wise): $\lambda A[i,j] = \lambda \cdot A[i,j]$

Vectors

Notation: $v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_d \end{pmatrix} \in \mathbb{R}^d$ column vector!

$\hookrightarrow$ can be seen as a matrix of size $d \times 1$.

Remark: row vectors = column vectors in Python. This is the case for 1D arrays.

If we want to be more specific we add another dimension to our array:

try creating an array, then reshape it.

Operations:

- dot product: $u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_d v_d$

Remark: A matrix $B \in \mathbb{R}^{n \times k}$ can be written as a collection of column vectors:

$$B = \begin{bmatrix} B[1,:] & B[2,:] & \dots & B[k,:] \end{bmatrix}$$

$\uparrow$ size $(n \times 1)$.

- We can define matrix multiplication:

$$AB[i,j] = \underbrace{A[i,:]}_{i\text{th-row of } A} \cdot \underbrace{B[:,j]}_{j\text{-th column of } B}$$

- We can define matrix vector multiplication:

$$A \in \mathbb{R}^{n \times k} \quad \begin{matrix} \nearrow \\ A \end{matrix} \quad \begin{matrix} \uparrow \\ v \end{matrix} \quad \begin{matrix} v \in \mathbb{R}^{k \times 1} \end{matrix} \quad = \quad A[:,1]v_1 + A[:,2]v_2 + \dots + A[:,k]v_k \in \mathbb{R}^{n \times 1}$$

$$Av[i,j] = A[i,:] \underbrace{v[:,j]}_{v_j}$$

Remark: Computational costs! $(n \cdot k)$ multiplications and $n \cdot (k-1)$ additions.

More operations and special matrices

- Transposing: We switch rows and columns. That is:

notation $\rightarrow B^T[i,j] = B[j,i]$.

If $B = B^T$ we say B is symmetric.

- Can we find a matrix $C \in \mathbb{R}^{n \times n}$ s.t for any matrix $A \in \mathbb{R}^{n \times n}$, $AC = A$?
(& $CA = A$)

Yes! We call this the identity matrix I :

$$I = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & 1 \end{pmatrix} \rightarrow \text{in Python: } \text{np.eye}(n).$$

- For some square matrices $A \in \mathbb{R}^{n \times n}$, their inverse exists:

$$A A^{-1} = A^{-1} A = I$$

Remark: In Python, numbering begins with 0:

$$v = \text{np.array}([1, 2, 3, 4, 5]) \Rightarrow v[0] = 1, v[1] = 2, \dots, v[4] = 5$$

or $v[-1]$ to access last element.

Lecture 2: Partition of sets

Motivation: Categorization. Neural networks!

What is an equivalence relation?

↳ We say a relation $\sim$ is equivalent if it's:

reflexive

$$h \sim h$$

symmetric

$$a \sim b \Rightarrow b \sim a$$

transitive

$$a \sim b \text{ \& \> } b \sim c \Rightarrow a \sim c$$

• Example:

Consider the rotation matrix $P_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ for some angle θ .

Collect all these rotations in a set:

$$\Lambda = \{P_\theta / \theta \in [0, 2\pi)\}$$

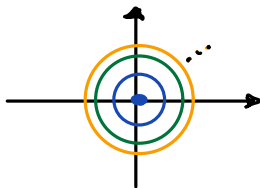
We define a relation $\hat{\sim}$ between two vectors $u, v \in \mathbb{R}^2$:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \hat{\sim} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \text{ if and only if there exists } \theta \text{ s.t. } \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = P_\theta \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

i.e. we can obtain one vector by rotating another

How do these equivalent classes look like?

Answer: concentric circles



We have obtained a non-intersecting partition of our set (in this case $\mathbb{R}^2$).

Linear mappings

We call $B: \mathbb{R}^k \rightarrow \mathbb{R}^n$ a linear mapping if

- For all $u, v \in \mathbb{R}^k$: $B(u+v) = Bu + Bv$

- For all $\lambda \in \mathbb{R}$: $B(\lambda u) = \lambda Bu$

- Example: matrix multiplication!

$$B: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \underbrace{\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}}_{\in \mathbb{R}^{3 \times 2}} \begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \end{pmatrix} + y \begin{pmatrix} b_{12} \\ b_{22} \\ b_{32} \end{pmatrix}$$

- Can we see this property of linearity visually?

Answer: Yes! $B(u+v) = B(u) + B(v)$ and $B(\lambda u) = \lambda B(u)$ imply

$$B(0) = B(-v+v) = B(-v) + B(v) = -B(v) + B(v) = 0$$

That is linear maps don't shift or change the origin!

- Can we obtain a matrix for every linear transformation?

Answer: Yes!

If $T: \mathbb{R}^k \rightarrow \mathbb{R}^n$ is a linear transformation, then

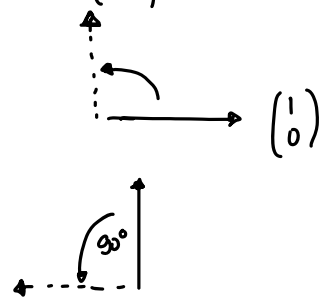
$T\underline{x}$ for $\underline{x} \in \mathbb{R}^k$ is equivalent to

$$\begin{pmatrix} T(\underline{e}_1) & T(\underline{e}_2) & \dots & T(\underline{e}_k) \end{pmatrix} \underline{x}$$

↑
our basis vectors

Example 90° rotation in $\mathbb{R}^2$:

let's see what this does to $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \xrightarrow{90^\circ} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \xrightarrow{90^\circ} \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$


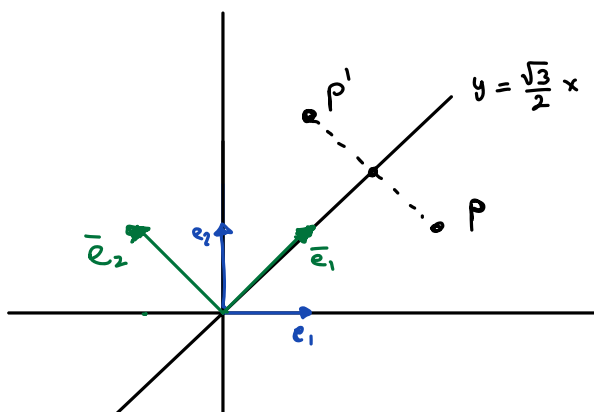
Then the corresponding matrix for this linear transformation is:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Change of basis

We saw that the matrix we choose to represent our linear transformation depends on the basis we choose. This can help us obtain a better looking matrix.

Example: Reflection to the line $y = \frac{\sqrt{3}}{2}x$ in $\mathbb{R}^2$



It can be shown that such a matrix in our standard basis has the form:

$$\frac{1}{1 + \left(\frac{\sqrt{3}}{2}\right)^2} \begin{pmatrix} 1 - \left(\frac{\sqrt{3}}{2}\right)^2 & 2\left(\frac{\sqrt{3}}{2}\right) \\ 2\left(\frac{\sqrt{3}}{2}\right) & \left(\frac{\sqrt{3}}{2}\right)^2 - 1 \end{pmatrix}$$

$$= \frac{4}{7} \cdot \begin{pmatrix} 1/4 & \sqrt{3} \\ \sqrt{3} & -1/4 \end{pmatrix}$$

Let's choose different basis!

$$\bar{e}_1 = \begin{pmatrix} 1 \\ \frac{\sqrt{3}}{2} \end{pmatrix}, \bar{e}_2 = \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ 1 \end{pmatrix} \Rightarrow$$

$$\bar{e}_1 = \begin{pmatrix} 1 \\ \frac{\sqrt{3}}{2} \end{pmatrix} = 1 \cdot e_1 + \frac{\sqrt{3}}{2} e_2$$

but now we write it using $\bar{e}_1$ and $\bar{e}_2$:

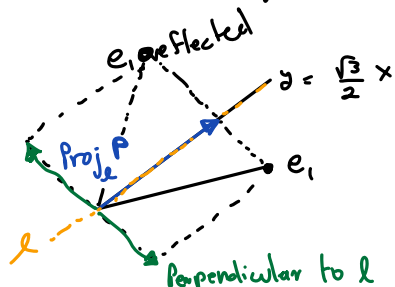
$$\bar{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}_{[\bar{e}_1, \bar{e}_2]} = 1 \cdot \bar{e}_1 + 0 \cdot \bar{e}_2$$

Reflection does not change $\bar{e}_1$: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
(written w.r.t. $\bar{e}_1, \bar{e}_2$)

$\bar{e}_2$ is now perpendicular to the line $y = \frac{\sqrt{3}}{2}x$,

thus its reflection is $-\bar{e}_2$: $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$
(written w.r.t. $\bar{e}_1, \bar{e}_2$)

Remark: Another way of computing the matrix with the original basis:



We know: $\text{Proj}_l e_1 = \frac{e_1 \cdot l}{|l|^2} l$

$$\text{Take } l = \begin{pmatrix} 1 \\ \frac{\sqrt{3}}{2} \end{pmatrix}$$

$$= \frac{\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ \frac{\sqrt{3}}{2} \end{bmatrix}}{1 + \left(\frac{\sqrt{3}}{2}\right)^2} \begin{bmatrix} 1 \\ \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$= \frac{4}{7} \begin{bmatrix} 1 \\ \frac{\sqrt{3}}{2} \end{bmatrix}$$

Since for any vector v , we have the decomposition $v = \text{proj}_L v + (\text{proj}_L v)^\perp$

$$\Rightarrow (\text{proj}_L e_1)^\perp = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 4/7 \\ \frac{2\sqrt{3}}{7} \end{pmatrix} = \begin{pmatrix} 3/7 \\ -\frac{2\sqrt{3}}{7} \end{pmatrix}$$

Then, the reflection of e_1 w.r.t L is given by

$$\text{proj}_L e_1 - (\text{proj}_L e_1)^\perp = \begin{pmatrix} 4/7 \\ \frac{2\sqrt{3}}{7} \end{pmatrix} - \begin{pmatrix} 3/7 \\ -\frac{2\sqrt{3}}{7} \end{pmatrix} = \begin{pmatrix} 1/7 \\ \frac{4\sqrt{3}}{7} \end{pmatrix}$$

Similarly for e_2 , we can obtain that its reflection is

$$\begin{pmatrix} \frac{4\sqrt{3}}{7} \\ -1/7 \end{pmatrix}.$$

Powers of a matrix

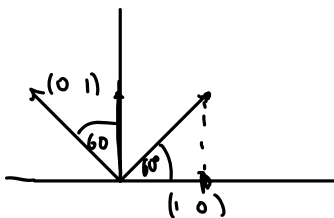
example:

• Matrix that flips 1st and 3rd component in $\mathbb{R}^3$:

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \Rightarrow A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \dots A^k = \begin{cases} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} & k \text{ even} \\ \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} & k \text{ odd} \end{cases}$$

• 60° rotation:

$$A = \begin{pmatrix} \cos(60^\circ) & -\sin(60^\circ) \\ \sin(60^\circ) & \cos(60^\circ) \end{pmatrix} = \begin{pmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix}$$



After 6 rotations, we should go back to the beginning:

$$A^6 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$